Casey Perdue

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Department of Mathematics
HSS 3072

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Academic Interests

I'm broadly interested in algebraic number theory and representation theory. More specifically, I'm interested in studying mod p representations of p-adic reductive groups and computing their cohomology groups, possibly arising from computations involving Lie algebra cohomology, as well as understanding Hecke algebras associated to Serre weights of such p-adic reductive groups.

Education

UC San Diego

- Ph.D., Mathematics (expected graduation: June 2027).
- M.A., Mathematics (2024).
- Advisor: Claus Sorensen.

UC Berkeley

- B.A., Mathematics, Highest Major Honors and High Distinction in General Scholarship (2021).
- GPA: 3.91 of 4.0.
- Major GPA: 3.97 of 4.0.

Publications and Preprints

2. **On the saturability of p-adic Lie groups**, preprint. (Link to preprint.)
1. **Higher mod p Satake transforms: explicit formulas and images for** GL_2 (with Hargun Bhatia, Yashi Jain, and Ben Savoie), preprint. (Link to preprint.)

Talks

4. *Higher mod p Satake transforms: Explicit formulas and images for GL_2.* Number Theory Seminar, UC San Diego, October 29, 2026.
3. *What is a p-adic Lie group?.* The San Marcos Informal Mathematics In-Person Colloquium, CSU San Marcos, October 6, 2026.
2. *On the saturability of p-adic Lie groups.* Advancement to PhD Candidacy, UC San Diego, February 2, 2026.
1. *Explicit generalized mod p Satake transforms.* Number Theory Seminar, UC San Diego, April 16, 2025.

[1] Updated October 15, 2026

Poster Presentations

2. *On the saturability of p-adic Lie groups.* The Australian Direction, Salt Lake City, Utah, August 11, 2026.

1. *On an inflation map for p-valued p-adic Lie groups.* Mathematical Congress of the Americas, Miami, Florida, July 21, 2025. (Received honorable mention in poster competition.)

Organization and Service

Mentorship

- Math Graduate Student Council Mentorship Program: In each of the academic years 2024–2025 and 2025–2026, I met monthly with a first year PhD student to help them acclimate to graduate school, and provided advice about various aspects of the program.
- Directed reading program Winter & Spring 2025: Met weekly with a promising undergraduate to mentor them while working through most of Ramakrishnan and Valenza's *Fourier Analysis on Number Fields*.
- Directed reading program Spring 2024: Met weekly with a promising undergraduate to mentor them while working through part of Fulton and Harris' *Representation Theory: A First Course*.

Professional service

- Organized weekly department teas during the academic year from January 2024 to June 2027.
- Graduate Affairs committee attendee: As a Senior TA in the 2026-2027 academic year, I attend twice-quarterly departmental Graduate Affairs committee meetings to provide recommendations about teaching policies, report any issues which arise for graduate student TAs, and broadly represent the graduate student body for matters related to teaching for the committee.
- Lead Teaching Assistant for Math 20C at UCSD: For four quarters (Fall 2023, Winter 2024, Fall 2024, and Winter 2025), I was one of two lead TAs for Math 20C (Calculus and Analytic Geometry for Science and Engineering). Lead TAs are selected by the department to assist in running large "superlecture" courses, which typically have over 1000 students across several lectures. Lead TAs' administrative duties include creating worksheets to be used in discussion sections, managing a team of graduate TAs and undergraduate tutors/readers, ensuring that exams are proctored, scanned, and graded promptly, and helping administer course policies.
- GPSA Representative: I volunteered to be a representative for the math department in the Graduate and Professional Student Association (GPSA) at UC San Diego for the 2022-2023 and 2023-2024 academic years. This involved attending biweekly meetings, voting on resolutions, and representing the interests of graduate students in my department. I also regularly reported back to my department with information presented at GPSA meetings.

Seminars

- Fall 2025: Co-organizer of seminar on *Cohomology of p-adic Chevalley groups*, with Hargun Bhatia.
- Winter 2025: Co-organizer of seminar on representations of p-adic groups, with Hargun Bhatia.
- Summer 2024: Organizer of seminar on the Langlands program.

Teaching Experience

UC San Diego

- Fall 2026: Senior TA for Math 500. In this role, I and my co-Senior TA ran the course (Math 500) where the incoming Math PhD cohort is trained on how to be successful TAs, under the supervision of a faculty member. We were assisted in this role by one Junior TA, who we selected after accepting applications and running interviews with all the competitive applicants.
- Summer Session II 2026: Primary course instructor for Math 20C (multivariable calculus). In this position, I designed a syllabus for my course, delivered a total 30 hours of lectures, and coordinated with my one TA for the content which would be covered in students' discussion sections.
- Spring 2026: Teaching Assistant for Math 220C. Unlike most TA positions, this did not involve running a discussion section as it was a graduate course, but instead involved holding extensive office hours, grading all homework, and grading most of the students' exam solutions.
- Winter 2026: Teaching Assistant for Math 220B. This was another graduate course.
- Fall 2025: Junior TA for Math 500. In this role, I assisted my co-Junior TA, as well as two Senior TAs, in running Math 500, where we trained the incoming Math PhD cohort to be successful TAs.
- Summer Session I 2025: Teaching Assistant for Math 142A.
- Spring 2025: Teaching Assistant for Math 109.
- Winter 2025: Lead Teaching Assistant for Math 20C. In this role, my co-lead TA and I created worksheets used in the discussion sections for over 1000 students and directly managed a team of 6 graduate TAs and 5 undergraduate tutors/readers, as well as running one discussion section each ourselves.
- Fall 2024: Lead Teaching Assistant for Math 20C. This quarter, my co-lead TA and I created worksheets for over 1250 students, and managed a team of 11 graduate TAs and 10 undergraduate tutors/readers, as well as running one discussion section each ourselves.
- Summer Session II 2024: Teaching Assistant for Math 10C.
- Spring 2024: Teaching Assistant for Math 109.
- Winter 2024: Lead Teaching Assistant for Math 20C. This quarter, there were over 950 students, and a team of 7 graduate TAs and 4 undergraduate tutors/readers.
- Fall 2023: Lead Teaching Assistant for Math 20C. This quarter, there were over 1050 students, and a team of 8 graduate TAs and 5 undergraduate tutors/readers.
- Summer Session I 2023: Teaching Assistant for Math 103A.
- Spring 2023: Teaching Assistant for Math 20E.
- Winter 2023: Teaching Assistant for Math 240B. This was another graduate course.
- Fall 2022: Teaching Assistant for Math 10B.

UC Berkeley

- Spring 2022: Reader for Math 143 (Elementary Algebraic Geometry).
- Fall 2021: Reader for Math 202A (Graduate Topology and Analysis) and Math 142A (Elementary Algebraic Topology).

- July-August 2021: Support Position for Pre-Calculus Essentials. In this position, I helped create and write solutions for daily worksheets and homework problems in the pre-calculus summer course at UC Berkeley. I also clarified topics for students by answering any questions they had during the course.
- Summer 2021: Reader for Math 115 (Intro to Number Theory).
- Spring 2021: Reader for Math 115 (Intro to Number Theory).
- Fall 2021: Reader for Math 135 (Intro to Set Theory).

Awards & Grants

- UC San Diego James B. Ax Graduate Fellowship ($12,000, disbursed 2022-2026).
- UC San Diego SHORE Fellowship (2022). Given to outstanding incoming graduate students at UC San Diego, this fellowship guaranteed on-campus housing for graduate students for up to six years.
- Dorothea Klumpke Roberts Prize (2021). Awarded to UC Berkeley seniors who have demonstrated truly exceptional scholarship in mathematics. In 2021, awarded to only 6 students.
- UC Berkeley Dean's List (Fall 2018, Fall 2019, Fall 2020). Awarded to top 10% of undergraduates in the College of Letters and Sciences who took at least 13 graded units each semester.